

Total No. of Questions : 10]

SEAT No. :

P3067

[5059]-548

[Total No. of Pages :4

B.E.(Mechanical)

FINITE ELEMENT ANALYSIS

(2012 Course)(Semester-II)(Elective-IV)(402050B)

Time :2½Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Draw suitable neat diagrams, wherever necessary.*
- 2) *Figures to the right indicate full marks.*
- 3) *Use of electronic pocket calculator is allowed.*
- 4) *Assume suitable data, if required.*

Q1) a) Explain step by step procedure for FEA and comment on convergence based on elemental size. [6]

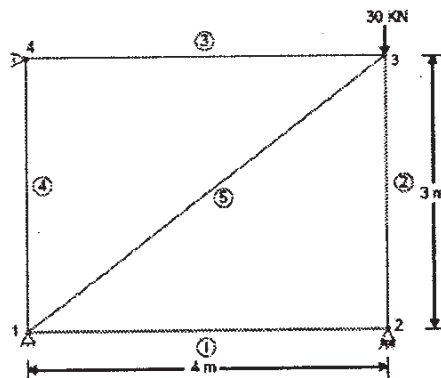
b) Explain concept of Plane Stress with appropriate example. [4]

OR

Q2) a) Write down the difference between Weighted Residual Method and Weak Formulations. [6]

b) Explain LST (Linear Strain Triangle Element) Element. [4]

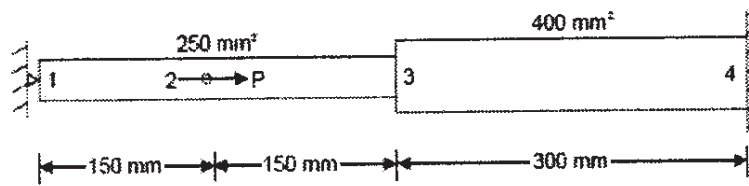
Q3) Determine the forces in the members of the truss shown in Fig. Take $E=200\text{GPa}$. $A=2000\text{mm}^2$. [10]



OR

P.T.O.

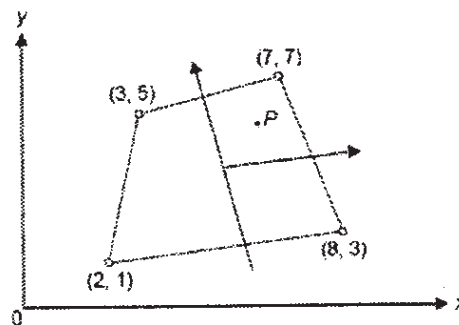
- Q4) a)** Determine the nodal displacement, element stresses and support reactions of the axially loaded bar as shown in Fig. Take $E=200\text{GPa}$ and $P=30\text{kN}$ [6]



- b) Write a note on Lagrange interpolation functions used in FEA formulations. [4]

- Q5) a)** Write a note on isoparametric formulations and how the geometric as well as field variable variation is taken into account? [6]

- b) Determine the Cartesian coordinate of the point P ($\xi=0.5, \eta=0.6$) shown in Fig. [4]



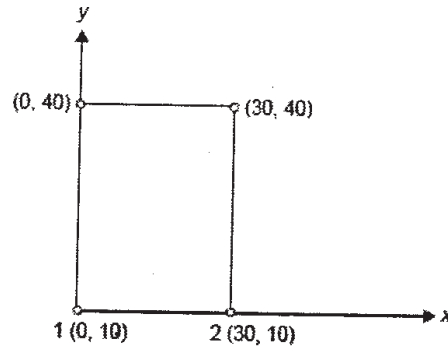
- c) Write short notes on [4]
- Uniqueness of mapping of isoparametric elements.
 - Jacobian matrix
- d) State and explain the three basic laws on which isoparametric concept is developed. [4]

OR

- Q6) a)** Write short notes on [8]

- Uniqueness of mapping of isoparametric elements.
- Jacobian matrix

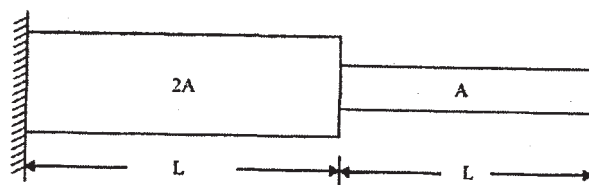
- b) For the element shown in Fig, assemble Jacobian matrix and strain displacement matrix for the Gaussian point(0.7,0.5). [10]



- Q7)** a) Write down governing equation of steady state heat transfer and also write down elemental stiffness matrix and compare with Bar element. [6]
- b) Consider a brick wall of thickness 0.6m, $k=0.75 \text{ W/m}^\circ\text{K}$. The inner surface is at 15°C and the outer surface is exposed to cold air at -15°C . The heat transfer coefficient associated with the outside surface is $40 \text{ W/m}^2\text{K}$. Determine the steady state temperature distribution within the wall and also the heat flux through the wall. Use two elements and obtain the solution. [10]

OR

- Q8)** a) Heat is generated in a large plate($K=0.5 \text{ W/m}^\circ\text{C}$) at the rate of 2000 W/m^3 . The plate is 10 cm thick. Outside surface of the plate is exposed to ambient air at 30°C with a convective heat transfer coefficient of $40 \text{ W/m}^2\text{ }^\circ\text{C}$. Determine the temperature distribution in the wall. [10]
- b) Derive FEA stiffness matrix for Pin Fin Heat Transfer problem. [6]
- Q9)** a) Write down Consistent Mass and Lumped Mass Matrix for [6]
- Bar Element
 - Plane Stress Element
- b) Find the natural frequencies of longitudinal vibrations of the same stepped shaft of areas $A=1200 \text{ mm}^2$ and $2A=2500 \text{ mm}^2$ and of equal lengths ($L=1 \text{ m}$), when it is constrained at one end, as shown below. [10]



OR

Q10)a) Explain difference between consistent and lumped mass matrix technique for modal analysis of structure. [6]

b) Find the natural frequencies of longitudinal vibrations of the unconstrained stepped shaft of areas A and $2A$ and of equal lengths(L), as shown below. [10]

